

SONiX 32-Bit Cortex-M0 Micro-Controller

DMA ADC Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/27	1. First version.

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1 OVERVIEW

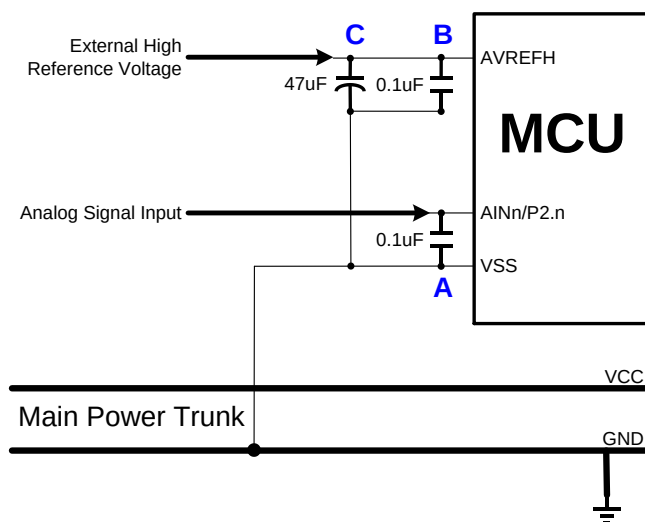
This document provides our customers the C program examples of ADC in DMA mode.

2 FEATURE

- Analog to Digital Converter function with DMA transfer.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- The ADC input signal must be through a 0.1uF capacitor. The 0.1uF capacitor is set between ADC input pin and VSS pin, and must be on the side of the ADC input pin as possible. Don't connect the capacitor's ground pin to ground plain directly, and must be through VSS pin. The capacitor can reduce the power noise effective coupled with the analog signal.



4 EXAMPLE CODE

4.1 Initialization

Initial ADC and DMA function. User can modify the settings of ADC and DMA in these function.

```
72 // Initial ADC
73 ADC_FuncInit(ADC_DIV32, ADC_12BIT, Multiple_Channel, Continuous_Mode);
74
75 // Init DMA
76 DMAADCInit();
```

For example: Set ADC high reference voltage source as Internal VDD.

ADC resolution as 12 bit.

ADC conversion time = $1 / (\text{ADC MCLK}/32) * 16 \text{ sec.}$

```
72 // Initial ADC
73 ADC_FuncInit(ADC_DIV32, ADC_12BIT, Multiple_Channel, Continuous_Mode);

298 void ADC_FuncInit(uint8_t bPCLKDiv, uint8_t bADCLen, uint8_t bCHMode, uint8_t bSCMode)
299 {
300     __ADC_ENABLE_HCLK; //Enables HCLK for ADC
301
302     SN_ADC->ADM_b.AVREFHSEL = ADC_AVREFHSEL_INTERNAL; //Set ADC high reference voltage source from internal reference
303     SN_ADC->ADM_b.VHS = ADC_VHS_INTERNAL_4P5V; //Set ADC high reference voltage source as Internal 4.5V
304
305     SN_ADC->ADM_b.GCHS = ADC_GCHS_EN; //Enable ADC global channel
306
307     SN_ADC->ADM_b.ADLEN = bADCLen; //Set ADC resolution = 12-bit
308
309     SN_ADC->ADM_b.ADCKS = bPCLKDiv; //ADC_CLK = ADC_PCLK/32
310
311     SN_ADC->CONVCTRL_b.SCMODE = bSCMode; //Set mode
312
313     SN_ADC->CONVCTRL_b.CH = bCHMode; //Set converting channel
314
315     SN_ADC->ADM_b.ADENB = ADC_ADENB_EN; //Enable ADC
316
317     UT_DelayNx10us(10); //Delay 100us
318
319     SN_ADC->ADM1_b.ACS = ENABLE; //Calibration start
320     while (SN_ADC->ADM1_b.ACS == 1);
321
322     SN_ADC->ADM1_b.CALIVALENB = ENABLE; //ADC conversion with calibration value
323 }
```

The setting of DMA should refer to the table in the datasheet.

```

96 void DMAADCInit(void)
97 {
98     DMA_InitSt stDMACH_Init;
99
100     // Set DMA CH0 as P(ADC) to M(SRAM) mode
101     // Source : Peripheral ADC, Fixed, 8bit, ADC, Burst 4
102     stDMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_FIX;
103     stDMACH_Init.b_SrcMode     = DMA_SRC_PERIPHERAL;
104     stDMACH_Init.b_SrcWidth    = DMA_SRC_WIDTH_16BIT;
105     stDMACH_Init.b_SrcReqSel   = DMA_SRCRS_ADC;
106     stDMACH_Init.b_BurstSize   = DMA_BURST_4;
107     // Destination : Memory, Increment, 8bit, no request
108     stDMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_INC;
109     stDMACH_Init.b_DstMode     = DMA_DST_MEMORY;
110     stDMACH_Init.b_DstWidth    = DMA_DST_WIDTH_16BIT;
111     stDMACH_Init.b_DstReqSel   = DMA_DSTRS_NONE;
112     // Channel : LV0, Enable TC/ABT interrupt
113     stDMACH_Init.b_Priority    = DMA_CHPRI_LV0;
114     stDMACH_Init.b_IntTCEn     = DMA_INT_TC_MSK_DIS;
115     stDMACH_Init.b_IntABTEn    = DMA_INT_ABT_MSK_DIS;
116
117     // Initial DMA
118     DMA_Init(&stDMACH_Init, eDMA_CH0);
119 }

```

4.2 DMA ADC function enable

DMA request will be issued when the ADC FIFO is greater than the threshold. The DMA TC flag and interrupt will be set when all data transfers are completed.

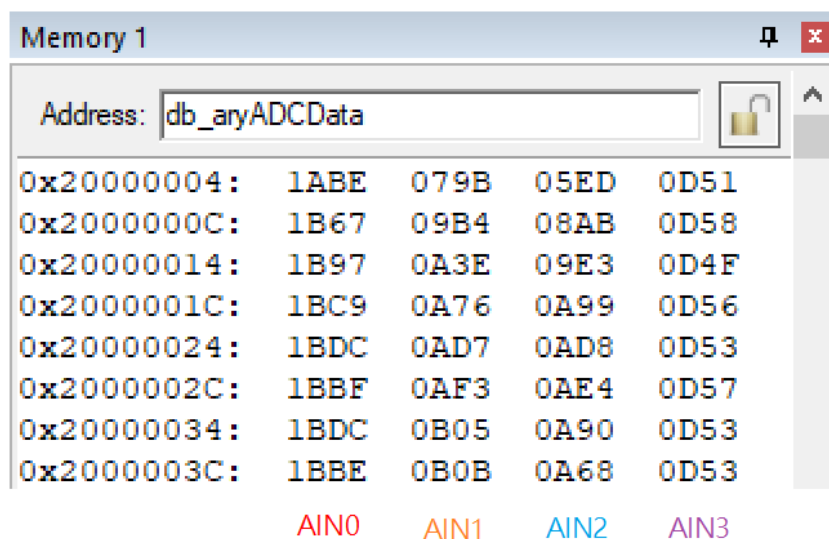
```

78 // Start DMA ADC
79 DMAADCStart();
80 ADC_DMA_Start(ADC_CHS_AIN0|ADC_CHS_AIN1|ADC_CHS_AIN2|ADC_CHS_AIN3, ADC_DMA_FIFO_TH_3, DMA_TEST_SIZE);
81
82 // Wait for DMA TC interrupt flag
83 while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);
84
85 while(1); // Pass
86 }

```

5 RESULT

The ADC conversion result will be moved to SRAM by DMA. The results can be observed through Watch window or Memory window.



0x20000004:	1ABE	079B	05ED	0D51
0x2000000C:	1B67	09B4	08AB	0D58
0x20000014:	1B97	0A3E	09E3	0D4F
0x2000001C:	1BC9	0A76	0A99	0D56
0x20000024:	1BDC	0AD7	0AD8	0D53
0x2000002C:	1BBF	0AF3	0AE4	0D57
0x20000034:	1BDC	0B05	0A90	0D53
0x2000003C:	1BBE	0B0B	0A68	0D53

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